Absolutely. Let us start from the **foundation** and make the whole thing clear.

Right now your confusion is natural, because in **machine learning / deep learning for time series**, the first question is always:

"What exactly am I trying to predict?"

That decision comes **before** selecting Random Forest, XGBoost, LSTM, or anything else.

---

1. First understand your data in the simplest way

Your IMF dataset contains, for each **country** and **year**, many macroeconomic variables such as:

- GDP growth
- inflation
- government debt
- fiscal balance
- export growth
- import growth
- investment
- savings
- unemployment
- current account balance

So for one country, the data looks conceptually like this:

Country	Year	GDP Growth	Inflation	Debt	Investment	Exports	Imports	Fiscal Balance	...
India	2018	6.5	3.9	69	31	8	11	-6.2	...
India	2019	4.0	3.7	72	30	2	4	-7.0	...
India	2020	-5.8	6.2	88	28	-10	-16	-9.5	...
India	2021	9.1	5.5	85	29	12	15	-6.8	...

So the natural question becomes:

Can last year's macroeconomic conditions help us predict next year's economic outcome?

That is where ML and deep learning enter.

---

2. What do we predict?

In your case, the best thing to predict is:

Target variable = GDP growth

Suggested name:
`gdp_growth`

Why GDP growth?

Because GDP growth is:

- one of the most important macroeconomic outcomes
- central to policy, planning, and development
- influenced by many other variables in your dataset
- suitable for regression-based ML and time-series forecasting

So your paper becomes:

> "Using past macroeconomic indicators to predict future GDP growth."

That is clear, logical, and academically valid.

---

3. Why not predict everything?

Because ML works best when the research question is focused.

If you try to predict:

- inflation
- debt
- exports
- imports

- unemployment
- GDP growth

all together from the beginning, the work becomes scattered.

A publishable paper usually has:

- **one main target variable**
- several predictor variables

So:

Main outcome to predict:

`gdp_growth`

Predictor variables:

- `inflation_cpi`
- `current_account_gdp`
- `investment_gdp`
- `gov_debt_gdp`
- `fiscal_balance_gdp`
- `export_volume_growth`
- `import_volume_growth`
- maybe `savings_gdp`
- maybe `unemployment_rate` in robustness analysis

---

4. What does "predict GDP growth" actually mean?

It means:

> Use information from previous years to estimate what GDP growth will be in the next year.

For example:

Can 2020 values of:

- inflation
- debt
- investment
- exports
- imports
- fiscal balance

help predict **GDP growth in 2021**?

That is the entire idea.

---

5. Very important concept: predictors must come from the past

This is the most important rule in time series and forecasting.

If you want to predict GDP growth in year t, you should use variables from year $t-1$, sometimes also $t-2$, $t-3$.

Why?

Because in real life, we do not know the future when making a forecast.

So this is wrong:

- using 2021 inflation to predict 2021 GDP growth in a forecasting setup

This is better:

- using 2020 inflation to predict 2021 GDP growth

So the structure becomes:

$$GDP\ Growth_t = f(Inflation_{t-1}, Debt_{t-1}, Investment_{t-1}, Exports_{t-1}, Imports_{t-1}, FiscalBalance_{t-1}, GDPGrowth_{t-1})$$

That is exactly what we feed into ML models.

---

6. So what are the X and y in machine learning here?

In ML language:

- **y** = target variable = what we want to predict
- **X** = predictor variables = what we use to predict

So in your project:

y

`gdp_growth_t`

X

Lagged macro variables such as:

- `gdp_growth_t-1`
- `inflation_cpi_t-1`
- `gov_debt_gdp_t-1`
- `investment_gdp_t-1`
- `fiscal_balance_gdp_t-1`
- `export_volume_growth_t-1`
- `import_volume_growth_t-1`
- `current_account_gdp_t-1`

---

7. Why do we include lagged GDP growth itself?

Because economies show **momentum and persistence**.

If a country grows strongly in one year, often that affects the next year too.

So `lagged GDP growth` is usually a strong predictor.

This is not cheating. This is standard time-series logic.

---

8. Why do we include the other macro variables?

Because they may influence future growth.

Inflation

High inflation may reduce stability and growth.

Investment

Higher investment usually supports future production and growth.

Government debt

Too much debt may slow growth or reduce fiscal space.

Fiscal balance

Persistent deficits may create pressure; moderate deficits may also support growth during downturns.

Export growth

May indicate external demand and competitiveness.

Import growth

May indicate domestic demand or capital goods imports.

Current account balance

Shows external sustainability or vulnerability.

So ML helps answer:

> Which of these past macro indicators are most useful for predicting next-year GDP growth?

---

9. Then what is the role of ML?

ML is not magic. It is simply a more flexible way to model the relationship between X and y.

Traditional regression says:

- assume mostly linear relationships

ML says:

- let the data show nonlinear patterns
- allow interactions automatically
- improve predictive performance

For example, regression may assume:

- 1 unit inflation increase causes the same effect everywhere

But ML may discover:

- low inflation has little effect
- very high inflation sharply reduces growth
- debt is only harmful when inflation is also high
- imports help growth only when investment is also strong

This is why ML is valuable.

---

10. Then where does deep learning come in?

Deep learning, especially **LSTM**, is useful when we want the model to learn from a **sequence of past years**, not just one lag.

For example:

Instead of only using 2020 values to predict 2021 GDP growth, LSTM can use:

- 2018
- 2019
- 2020

to predict:

- 2021

So deep learning asks:

> Can a longer historical sequence improve prediction?

That is why it is attractive for time-series research.

---

11. The full logic of your project in one sentence

Your project can be framed as:

> "To predict next-year GDP growth for countries using past macroeconomic indicators, and to compare econometric, machine learning, and deep learning methods in capturing temporal and nonlinear macroeconomic relationships."

That is a clean Scopus-level idea.

---

12. Step-by-step: what to do, how to do it, and why to do it

Now I will give you the full process from basic level.

Step 1: Decide the research question

What to do

Choose a clear question.

Recommended question

Can past macroeconomic indicators predict future GDP growth across countries?

Why

Without a clear question, ML becomes random and directionless.

Step 2: Choose the target variable

What to do

Select one variable to predict.

Recommended choice

`gdp_growth`

Why

It is economically important and your other variables can logically explain it.

Step 3: Choose predictor variables

What to do

Select variables that may influence GDP growth.

Good predictors

- lagged GDP growth
- inflation
- debt
- investment
- fiscal balance
- export growth
- import growth
- current account balance

Why

These represent major macroeconomic channels.

Step 4: Convert the data into forecasting form using lags

What to do

Create lagged variables country-wise.

For each country:

- previous year inflation
- previous year debt
- previous year GDP growth
- previous year exports

- previous year imports

Example

Country	Year	GDP Growth_t	GDP Growth_t-1	Inflation_t-1	Debt_t-1	Imports_t-1
India	2021	9.1	-5.8	6.2	88	-16

Why

Because we predict the future from the past.

Step 5: Clean missing values

What to do

Check which variables have too many missing values.

Why

ML models need usable input rows. If a variable has too many missing observations, it may reduce the sample too much.

In your data

Unemployment has heavy missingness, so it should probably be:

- excluded from the main model
- included only in a smaller robustness model

Step 6: Split the data by time, not randomly

What to do

Use earlier years for training and later years for testing.

Example

- Train: 1995–2015
- Validation: 2016–2019
- Test: 2020–2023

Why

In time series, future data must not leak into the past. Random splitting is usually inappropriate.

Step 7: Build a simple baseline first

What to do

Start with a linear benchmark model such as:

- linear regression
- panel regression
- autoregressive baseline

Why

You need a basic reference point. Without a benchmark, you cannot prove ML improved anything.

Step 8: Apply machine learning models

What to do

Train models like:

- Random Forest Regressor
- XGBoost Regressor
- Elastic Net
- Support Vector Regression

Why

These models can capture:

- nonlinearities
- interactions
- complex predictor-response patterns

Step 9: Apply deep learning only after ML

What to do

Train an LSTM or GRU model using sequences of past years.

Example

Use past 3 years of macro data to predict the next year's GDP growth.

Why

Deep learning is more advanced and data-hungry. It should come after a strong baseline and ML framework.

Step 10: Compare all models

What to do

Evaluate performance using:

- RMSE
- MAE
- R^2
- MAPE if meaningful

Why

You need to show which method predicts best.

Step 11: Interpret the best model

What to do

Use feature importance or SHAP values.

Why

A paper should not say only:

- "XGBoost was best"

It should also say:

- “import growth, lagged GDP growth, inflation, and debt were the most influential predictors”

That makes it policy-relevant and publishable.

Step 12: Translate prediction results into economic meaning

What to do

Explain what the model means economically.

Example

If import growth strongly predicts GDP growth, then imports may reflect:

- capital goods demand
- industrial inputs
- domestic demand expansion

Why

Scopus papers need interpretation, not just algorithms.

---

13. What exact ML problem type is this?

This is a **supervised regression problem**.

Why?

Because:

- target variable `gdp_growth` is numeric
- we want to predict a continuous value

So this is **not classification**.

It is:

Time-aware panel regression / forecasting problem

---

14. What can be your research designs from this same dataset?

You actually have several options.

Option A. Predict GDP growth

This is the best starting point.

Target

`gdp_growth`

Predictors

all lagged macro variables

Best for

a first strong paper

Option B. Predict inflation

Possible, but less attractive than GDP growth for a general macro paper.

Target

`inflation_cpi`

Predictors

lagged debt, growth, imports, current account, etc.

Best for

a more specialized macro-stability paper

Option C. Predict crisis vs non-crisis regime

This is a classification problem.

Target

Create a binary variable:

- 1 = recession / low-growth year
- 0 = non-recession year

Why

This lets you use classification ML models.

But

This is a second-stage paper, not the first paper.

Option D. Forecast multiple macro indicators

Possible, but more complicated and less clean initially.

So for now:

Stay with Option A: predict GDP growth

---

15. Which variables should be used first?

Start simple. Do not use every variable immediately.

Main first model variables

- `lag_gdp_growth`
- `lag_inflation_cpi`
- `lag_current_account_gdp`
- `lag_investment_gdp`
- `lag_gov_debt_gdp`
- `lag_fiscal_balance_gdp`
- `lag_export_volume_growth`
- `lag_import_volume_growth`

These are enough for a serious first model.

---

16. What does the dataset look like after preparation?

After lagging, each row will mean:

> For a given country in a given year, predict current GDP growth using the previous year's macro conditions.

Example:

Country	Year	GDP Growth (target)	Lag GDP Growth	Lag Inflation	Lag Debt	Lag Investment	Lag Export Growth	Lag Import Growth
Brazil	2021	4.8	-3.3	3.2	88	18	-5	-8

So the model learns from many such rows across countries and years.

---

17. Then how is deep learning data formed?

For deep learning, especially LSTM, one row is replaced by a **sequence**.

Instead of:

- just 2020 values → predict 2021 GDP growth

you do:

- 2018 values
- 2019 values
- 2020 values

together → predict 2021 GDP growth

So the model sees a mini time-history.

This is useful because economic movement is often influenced by patterns over several years, not only one lag.

---

18. Why not jump directly to deep learning?

Because that is a common mistake.

Deep learning is not automatically better.

You should go in this order:

1. baseline regression
2. machine learning
3. deep learning

Why?

Because then your paper can say:

- regression gives baseline
- ML improves nonlinear prediction
- deep learning tests longer temporal memory

That makes the paper methodologically mature.

---

19. What can be an ideal final workflow for your paper?

Here is the cleanest structure.

Stage 1: Econometric baseline

Predict GDP growth using lagged macro variables with panel regression.

Stage 2: Machine learning prediction

Use the same lagged variables in:

- Random Forest
- XGBoost
- Elastic Net
- SVR

Stage 3: Deep learning prediction

Use 3-year or 5-year sequences in LSTM.

Stage 4: Explainable AI

Use SHAP on the best ML model.

Stage 5: Economic interpretation

Translate model findings into policy logic.

---

20. In very plain language: what are we really doing?

We are asking this:

> If I know how a country looked last year in terms of inflation, debt, investment, trade, and fiscal position, can I predict how fast that country will grow this year?

That is the simplest and most correct explanation of your project.

---

21. Why this becomes a good research project

Because it combines:

- economics
- time series logic
- machine learning
- deep learning
- forecasting
- interpretation

So it is not just “run XGBoost for no reason.”

It becomes a meaningful research pipeline.

---

22. Recommended final beginner-friendly plan

Here is the exact plan I recommend for you.

Phase 1: Start with one target

Predict `gdp_growth`.

Phase 2: Use only lag-1 predictors first

Use previous year macro variables.

Phase 3: Train simple ML models

- Linear Regression
- Random Forest
- XGBoost

Phase 4: Compare model accuracy

Use RMSE, MAE, R^2.

Phase 5: Interpret important features

Use feature importance / SHAP.

Phase 6: Then move to LSTM

Use past 3 years to predict the next year.

This is the most natural path.

---

23. One final conceptual diagram in words

Think of it like this:

Input:

Past macroeconomic condition of a country

Model:

Regression / Random Forest / XGBoost / LSTM

Output:

Predicted GDP growth next year

That is all.

---

24. My direct recommendation

For your paper, do this:

Predict:

next-year GDP growth

Use as predictors:

previous-year macroeconomic variables

Compare:

- Panel regression
- Random Forest
- XGBoost
- LSTM

Then interpret:

which variables matter most and whether nonlinear models outperform linear models

That is a strong, clean, publishable framework.

Next I can give you the **exact variable-wise design table** showing:

- dependent variable
- independent variables
- lag structure
- model type
- why each variable is included
- what hypothesis each one implies.